CBSE Class 10 Mathematics — Triangles — Solutions

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1. 1. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 2$ cm, $DB = 3$ cm, and $AE = 3$ cm. Find the length of EC.

Answer: (C)

Solution:

1. Apply Thales Theorem (Basic Proportionality Theorem) as $DE \parallel BC$.
2. The ratio $\frac{AD}{DB} = \frac{AE}{EC}$ must hold.
3. Substitute the given values: $\frac{2}{3} = \frac{3}{EC}$.
4. Solve for EC: $EC = \frac{3\times 3}{2} = 4.5$ cm.

Derivation:

$$\begin{aligned} \frac{AD}{DB} &= \frac{AE}{EC} \\ \implies \frac{2}{3} &= \frac{3}{EC} \\ \implies 2EC &= 9 \\ \implies EC &= 4.5 \end{aligned}$$

Triangles — Basic Proportionality Theorem

2. 2. If $\triangle ABC \sim \triangle PQR$ and $\frac{AB}{PQ} = \frac{1}{3}$, then the ratio of their areas $\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)}$ is:

Answer: (B)

Solution:

1. Recall the theorem: The ratio of areas of two similar triangles is equal to the square of the ratio of their corresponding sides.
2. Calculate the square of the given ratio: $(\frac{1}{3})^2$.
3. Result is $\frac{1}{9}$.

Derivation:

$$\frac{\text{Area}(ABC)}{\text{Area}(PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

Triangles — Areas of Similar Triangles

3. 3. In $\triangle ABC$, if $AB = 6\sqrt{3}$ cm, $AC = 12$ cm, and $BC = 6$ cm, then the measure of $\angle B$ is:

Answer: (D)

Solution:

1. Check if the sides satisfy the Converse of Pythagoras Theorem: $AB^2 + BC^2 = AC^2$.
2. Calculate squares: $(6\sqrt{3})^2 = 36 \times 3 = 108$, $6^2 = 36$, $12^2 = 144$.

3. Observe $108 + 36 = 144$. Since $AB^2 + BC^2 = AC^2$, the triangle is right-angled at B.

Derivation:

$$(6\sqrt{3})^2 + 6^2 = 108 + 36 = 144 = 12^2$$

Triangles — Pythagoras Theorem

4. 4. Two triangles are similar if their corresponding sides are in the same ratio. This criterion is known as:

Answer: (A)

Solution:

1. SSS (Side-Side-Side) similarity criterion states that if the sides of one triangle are proportional to the sides of another triangle, then the triangles are similar.

Derivation:

$$\text{If } \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}, \text{ then } \triangle ABC \sim \triangle PQR \text{ (SSS Similarity)}$$

Triangles — Criteria for Similarity

5. 5. The altitude of an equilateral triangle with side $2a$ is:

Answer: (C)

Solution:

1. In an equilateral triangle, the altitude bisects the base.
2. Using Pythagoras theorem in the half-triangle: $h^2 + a^2 = (2a)^2$.
3. Solve for h: $h^2 = 4a^2 - a^2 = 3a^2 \implies h = a\sqrt{3}$.

Derivation:

$$h = \sqrt{(2a)^2 - a^2} = \sqrt{4a^2 - a^2} = \sqrt{3a^2} = a\sqrt{3}$$

Triangles — Properties of Triangles

6. 6. Assertion (A): If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side. Reason (R): This is the Converse of Basic Proportionality Theorem.

Answer: (A)

Solution:

1. Identify the statement as the definition of the converse of Thales Theorem (BPT).
2. Confirm that the reason correctly identifies the theorem name.

Derivation:

$$A : \text{Converse of BPT holds true. } R : \text{Definition of Converse of BPT.}$$

Triangles — Basic Proportionality Theorem

7. **7.** Assertion (A): Two triangles are similar if their corresponding angles are equal. Reason (R): If two triangles are equiangular, they are similar by AAA similarity criterion.

Answer: (A)

Solution:

1. Verify the AAA similarity criterion.
2. Check if equality of angles leads to similarity.

Derivation:

$$A: \text{Angle-Angle-Angle Similarity Criterion } R: \text{AAA creates proportional sides.}$$

Triangles — Similarity of Triangles

8. **8.** Assertion (A): In $\triangle ABC$, if $DE \parallel BC$ such that $AD/DB = 1/2$ and $AE = 2$ cm, then $EC = 4$ cm. Reason (R): By Basic Proportionality Theorem, if $DE \parallel BC$, then $AD/DB = AE/EC$.

Answer: (A)

Solution:

1. Apply BPT: $1/2 = 2/EC$.
2. Solve for EC: $EC = 4$ cm.

Derivation:

$$\frac{AD}{DB} = \frac{AE}{EC}$$
$$\implies \frac{1}{2} = \frac{2}{EC}$$
$$\implies EC = 4$$

Triangles — Basic Proportionality Theorem

9. **9.** Assertion (A): All congruent triangles are similar. Reason (R): All similar triangles are congruent.

Answer: (C)

Solution:

1. Congruent triangles have same shape and size, so they are similar.
2. Similar triangles have same shape but not necessarily same size, so they are not always congruent.

Derivation:

$$\text{Congruence}$$
$$\implies \text{Similarity (True). Similarity} Rightarrow \text{Congruence (False).}$$

Triangles — Similarity vs Congruence

10. 10. Assertion (A): The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. Reason (R): This is the Area of Similar Triangles Theorem.

Answer: (A)

Solution:

1. Recall the theorem regarding areas of similar triangles.
2. Confirm that the statement matches the standard theorem.

Derivation:

$$\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{AC}{PR}\right)^2$$

Triangles — Areas of Similar Triangles

11. 11. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 2.4$ cm, $AE = 3.2$ cm, and $EC = 4.8$ cm. Find the length of DB.

Answer:

$$3.6 \text{ cm}$$

Solution:

1. Apply Basic Proportionality Theorem (Thales Theorem).
2. Substitute the given values and solve for DB.

Derivation:

$$\frac{AD}{DB} = \frac{AE}{EC}$$
$$\implies \frac{2.4}{DB} = \frac{3.2}{4.8}$$
$$\implies DB = \frac{2.4 \times 4.8}{3.2} = 3.6 \text{ cm}$$

Triangles — Basic Proportionality Theorem

12. 12. Two triangles $\triangle ABC$ and $\triangle DEF$ are similar. If $Area(\triangle ABC) = 64$ cm^2 and $Area(\triangle DEF) = 121$ cm^2, and $EF = 15.4$ cm, find the length of side BC.

Answer:

$$11.2 \text{ cm}$$

Solution:

1. Use the theorem that the ratio of areas of similar triangles equals the square of the ratio of their corresponding sides.
2. Substitute values and take the square root to find BC.

Derivation:

$$\frac{Area(ABC)}{Area(DEF)} = \left(\frac{BC}{EF}\right)^2$$
$$\implies \frac{64}{121} = \left(\frac{BC}{15.4}\right)^2$$
$$\implies \frac{8}{11} = \frac{BC}{15.4}$$
$$\implies BC = \frac{8 \times 15.4}{11} = 11.2 \text{ cm}$$

Triangles — Areas of Similar Triangles

13. **13.** A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

Answer:

$$42 \text{ m}$$

Solution:

1. Recognize that the triangles formed by the object and shadow are similar by AA similarity.
2. Set up the ratio of corresponding sides and solve for the height.

Derivation:

$$\frac{6}{h} = \frac{4}{28}$$
$$\implies h = \frac{6 \times 28}{4} = 6 \times 7 = 42 \text{ m}$$

Triangles — Similar Triangles

14. **14.** In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $\angle PST = \angle PRQ$. If $PS = 2$ cm, $SQ = 3$ cm, and $PT = 2.5$ cm, find TR.

Answer:

$$1.5 \text{ cm}$$

Solution:

1. Show $\triangle PST \sim \triangle PRQ$ by AA similarity criterion.
2. Write the ratio of sides and solve for TR.

Derivation:

$$\triangle PST \sim \triangle$$
$$\implies \frac{PS}{PR} = \frac{PT}{PQ} \text{ is incorrect; use } \frac{PS}{PR} = \frac{PT}{PQ} \text{ is not needed, use } \frac{PS}{PT} = \frac{PT}{PR} \text{ is not right; use } \frac{PS}{PR} =$$
$$\implies \frac{PS}{PT + TR} = \frac{P}{PS +}$$
$$\implies \frac{2}{2.5 + TR} =$$
$$\implies 10 = 6.25 + 2$$
$$\implies 3.75 = 2$$
$$\implies TR = 1$$

15. 15. Diagonals of a trapezium $ABCD$ with $AB \parallel DC$ intersect each other at the point O. If $AB = 2CD$, find the ratio of the areas of triangles $\triangle AOB$ and $\triangle COD$.

Answer:

$$4 : 1$$

Solution:

1. Prove $\triangle AOB \sim \triangle COD$ by AA similarity.
2. Use the ratio of areas theorem.

Derivation:

$$\begin{aligned} \triangle AOB &\sim \triangle COD \\ \implies \frac{Area(AOB)}{Area(COD)} &= \left(\frac{AB}{CD}\right)^2 = \left(\frac{2CD}{CD}\right)^2 = 2^2 = 4 \end{aligned}$$

Triangles — Areas of Similar Triangles

16. 16. In $\triangle ABC$, $DE \parallel BC$ such that $AD = (4x - 3)$ cm, $AE = (8x - 7)$ cm, $BD = (3x - 1)$ cm, and $EC = (5x - 3)$ cm. Find the value of x.

Answer:

$$x = 1$$

Solution:

1. Since $DE \parallel BC$, by Basic Proportionality Theorem (BPT), $\frac{AD}{DB} = \frac{AE}{EC}$.
2. Substitute the given values into the equation: $\frac{4x-3}{3x-1} = \frac{8x-7}{5x-3}$.
3. Cross-multiply to get: $(4x - 3)(5x - 3) = (8x - 7)(3x - 1)$.
4. Solve the resulting linear equation to find x.

Derivation:

$$\begin{aligned} 20x^2 - 12x - 15x + 9 &= 24x^2 - 8x - 21x + 7 \\ \implies 20x^2 - 27x + 9 &= 24x^2 - 29x + 7 \\ \implies 4x^2 - 2x - 2 &= 0 \\ \implies 2x^2 - x - 1 &= 0 \\ \implies (2x + 1)(x - 1) &= 0. \text{ Since } x > 1, x = 1. \end{aligned}$$

Triangles — Basic Proportionality Theorem

17. 17. Two poles of heights 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.

Answer:

$$13m$$

Solution:

1. Let $AB = 6$ m and $CD = 11$ m be the two poles. The distance $BD = 12$ m.

2. Draw a line from A parallel to BD meeting CD at E. Then $CE = CD - ED = 11 - 6 = 5$ m.
3. In right-angled triangle $\triangle AEC$, use the Pythagorean theorem to find the hypotenuse AC.
4. $AC^2 = AE^2 + CE^2$.

Derivation:

$$AE = BD = 12 \text{ m.} CE = 11 - 6 = 5 \text{ m.} AC = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13 \text{ m.}$$

Triangles — Pythagoras Theorem

18. **18.** In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $\angle PST = \angle PRQ$. Show that $\triangle PQR \sim \triangle PTS$.

Answer:

$$Proof$$

Solution:

1. Identify the common angle in both triangles: $\angle QPR = \angle TPS$ (Common).
2. Use the given condition $\angle PST = \angle PRQ$.
3. Conclude similarity by AA (Angle-Angle) criterion.

Derivation:

$$\text{In } \triangle PQR \text{ and } \triangle PTS : \angle P = \angle P \text{ (Common)}, \angle PST = \angle PRQ \text{ (Given). Therefore, } \triangle PQR \sim \triangle P$$

Triangles — Criteria for Similarity of Triangles

19. **19.** The areas of two similar triangles are 121 cm^2 and 64 cm^2 respectively. If the median of the first triangle is 12.1 cm, find the corresponding median of the other triangle.

Answer:

$$8.8cm$$

Solution:

1. Use the theorem: Ratio of areas of similar triangles is equal to the ratio of the squares of their corresponding medians.
2. Set up the equation: $\frac{Area_1}{Area_2} = \left(\frac{m_1}{m_2}\right)^2$.
3. Substitute values: $\frac{121}{64} = \left(\frac{12.1}{m_2}\right)^2$.
4. Take the square root and solve for m_2.

Derivation:

$$\frac{11}{8} = \frac{12.1}{m_2}$$
$$\implies m_2 = \frac{12.1 \times 8}{11} = 1.1 \times 8 = 8.8 \text{ cm.}$$

Triangles — Areas of Similar Triangles

20. **20.** In $\triangle ABC$, $AD \perp BC$ such that $AD^2 = BD \cdot CD$. Prove that $\angle BAC = 90°$.

Answer:

$$Proof$$

Solution:

1. Since $AD \perp BC$, we have two right triangles $\triangle ADB$ and $\triangle ADC$.
2. Express the sides using Pythagoras: $AB^2 = AD^2 + BD^2$ and $AC^2 = AD^2 + CD^2$.
3. Add them and use the given condition $AD^2 = BD \cdot CD$ to show $AB^2 + AC^2 = BC^2$.

Derivation:

$AB^2 + AC^2 = AD^2 + BD^2 + AD^2 + CD^2 = 2AD^2 + BD^2 + CD^2$. Substitute $AD^2 = BD \cdot CD : AB^2 + AC$

Triangles — Pythagoras Theorem

21. **21.** State and prove the Basic Proportionality Theorem (Thales Theorem).

Answer:

$Ifalineisdrawnparalleltoonesideofatriangletointersecttheothertwosidesindistinctpoints, theothe$

Solution:

1. Draw a triangle ABC with a line DE parallel to BC intersecting AB at D and AC at E.
2. Construct perpendiculars DM on AC and EN on AB. Join BE and CD.
3. Calculate the area of triangles ADE, BDE, and CDE.
4. Use the property that triangles on the same base and between the same parallels have equal areas.
5. Divide the ratio of areas to prove the theorem.

Derivation:

$$In \triangle ADE and \triangle BDE, \frac{ar(ADE)}{ar(BDE)} = \frac{\frac{1}{2} \cdot AD \cdot EN}{\frac{1}{2} \cdot BD \cdot EN} = \frac{AD}{BD}. Similarly, \frac{ar(ADE)}{ar(CDE)} = \frac{AE}{CE}. Sincear(BD$$

Triangles — Basic Proportionality Theorem

22. **22.** Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.

Answer:

$$\frac{ar(\triangle ABC)}{ar(\triangle PQR)} = (\frac{AB}{PQ})^2 = (\frac{BC}{QR})^2 = (\frac{AC}{PR})^2$$

Solution:

1. Draw two similar triangles ABC and PQR.
2. Draw altitudes AM $\perp$ BC and PN $\perp$ QR.
3. Express the ratio of areas as $\frac{\frac{1}{2} \cdot BC \cdot AM}{\frac{1}{2} \cdot QR \cdot PN}$.

4. Prove $\triangle ABM \sim \triangle PQN$ to show $\frac{AM}{PN} = \frac{AB}{PQ}$.
5. Substitute this ratio back into the area equation to get the square of the ratio.

Derivation:

$$\frac{ar(ABC)}{ar(PQR)} = \frac{BC \cdot AM}{QR \cdot PN} = \frac{BC}{QR} \cdot \frac{AM}{PN}.Since\triangle ABM \sim \triangle PQN, \frac{AM}{PN} = \frac{AB}{PQ}.Also\frac{AB}{PQ} = \frac{BC}{QR}(givensi$$

Triangles — Areas of Similar Triangles

23. 23. In $\triangle ABC$, $AD \perp BC$ such that $AD^2 = BD \cdot CD$. Prove that $\triangle ABC$ is a right-angled triangle at A.

Answer:

$$\angle BAC = 90^\circ$$

Solution:

1. Observe the triangle split into two smaller triangles $\triangle ABD$ and $\triangle ACD$.
2. Write the ratio $\frac{AD}{BD} = \frac{CD}{AD}$.
3. Include the angle $\angle ADB = \angle ADC = 90^\circ$.
4. Prove $\triangle ABD \sim \triangle CAD$ by SAS similarity.
5. Equate corresponding angles $\angle BAD = \angle ACD$ and $\angle CAD = \angle ABD$.
6. Use the angle sum property of $\triangle ABC$ to show $\angle BAC = 90^\circ$.

Derivation:

$$\implies \frac{AD}{BD} = \frac{CD}{AD}.In\triangle ABDand\triangle CAD, \angle ADB = \angle ADC = 90^\circ.BySASsimilarity, \triangle ABD \sim \triangle C$$

Triangles — Similarity of Triangles

24. 24. A vertical stick 2 m long casts a shadow 1 m long on the ground. At the same time, a tower casts a shadow 5 m long. Find the height of the tower using the properties of similar triangles.

Answer:

$$10meters$$

Solution:

1. Identify two triangles: one formed by the stick and its shadow, the other by the tower and its shadow.
2. State that at the same time, the angle of elevation of the sun is the same for both objects.
3. Identify that both triangles are right-angled and share an equal angle, thus they are similar by AA criterion.
4. Set up the proportion: $\frac{\text{Height of Stick}}{\text{Shadow of Stick}} = \frac{\text{Height of Tower}}{\text{Shadow of Tower}}$.

5. Substitute the given values and solve for the height of the tower.

Derivation:

$$Let height of tower be h. \triangle ABC \sim \triangle PQR (where ABC is stick triangle and PQR is tower triangle).The$$

Triangles — Applications of Similar Triangles